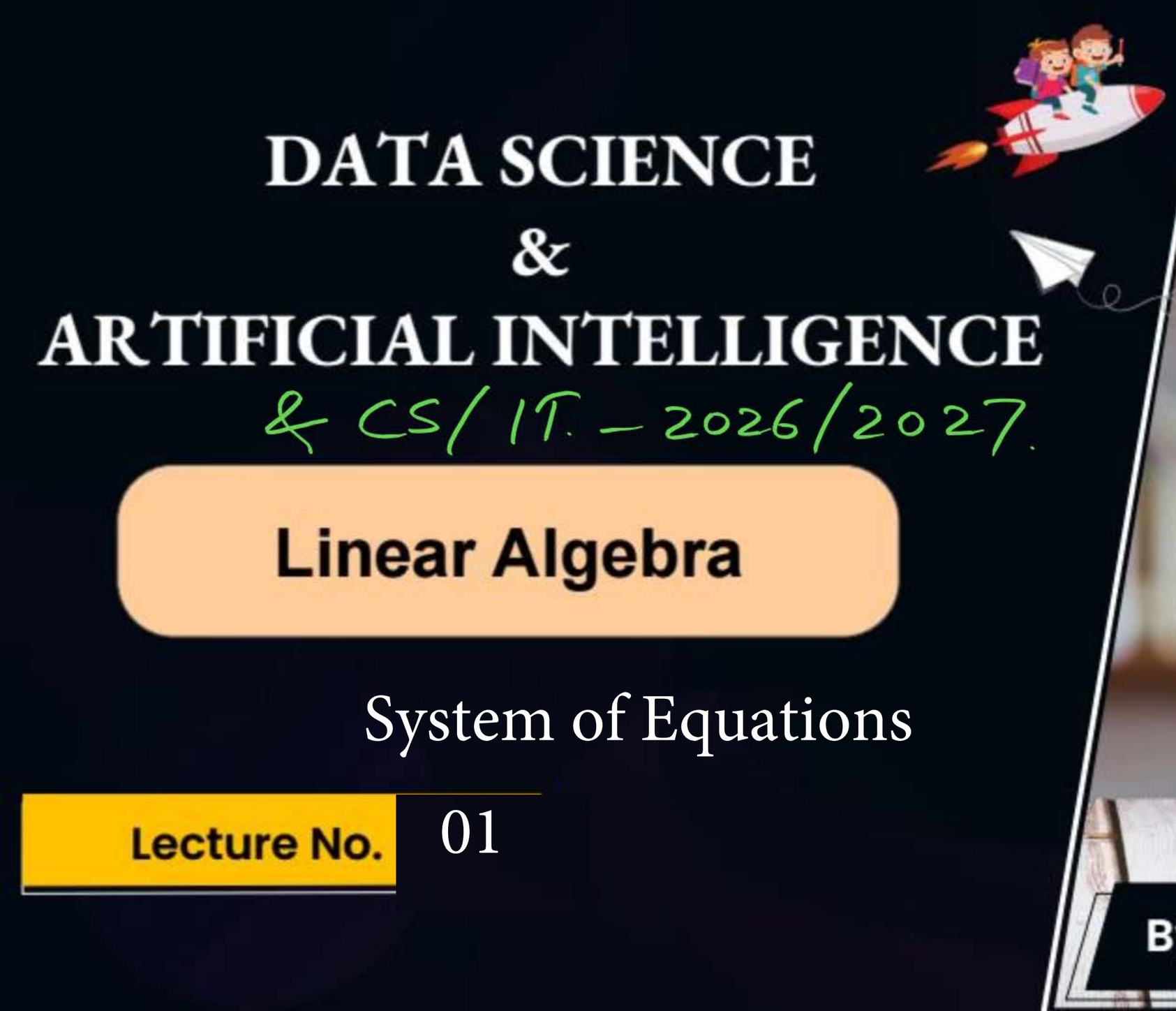
DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT. – 2026/2027.

Linear Algebra

System of Equations

Lecture No. 01



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Recap of previous lecture



Topic

1. ECHELON FORM
2. NATURE of VECTORS

Topics to be Covered



Topic

1. LD & LI VECTORS
2. Non Homogeneous System of Equations

Linear Combination of Vectors → (Linear Relationship b/n the vectors)

Let $X_1, X_2, X_3 \dots X_r$ are given vectors & let $K_1, K_2, K_3 \dots K_r$ are the scalars (Constant)

Then the Relationship of the type

$$K_1X_1 + K_2X_2 + K_3X_3 + \dots + K_rX_r = 0 \quad \text{---(1)}$$

is called L-combination of vectors.

Linearly Dependent vectors → vectors are called (LD) if Linear Relationship b/n them

Linearly Independent Vectors → " " " (LI) if there DNE any " " "

eg if $K_1X_1 + K_2X_2 + K_3X_3 + \dots + K_rX_r = 0$ exist only when $K_1 = K_2 = K_3 = \dots = K_r = 0$

then vectors are LI i.e $0X_1 + 0X_2 + 0X_3 + \dots + 0X_r = 0$



Methods to check Nature of vectors

Let $x_1, x_2, x_3 \ldots x_r$ are the given vectors then Construct a new Mat of the type $A = [x_1\ x_2\ x_3 \ldots x_r]$

General Method :- (RANK Method)

(i) If $S(A)$ = number of vectors $(r) \Rightarrow$ vectors are (L.I)

(ii) If $S(A) <$ " " " $\Rightarrow$ " " L.D

Tricky Method → Applicable only when A is Sq-Mat.

(i) If $|A| \neq 0 \Rightarrow$ vectors are (L.I)

(ii) If $|A| = 0 \Rightarrow$ " " L.D

(*) In Case of two vectors X_1 & X_2, No Need to follow General Method or Tricky Method, only follow Observation Method i.e

(i) if $X_1 = KX_2$ for any Non Zero Scalar K then vectors are LD

(ii) if $X_1 \neq KX_2$ for any K then vectors are LI

eg $X_1 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ 3 \end{bmatrix}$, $X_2 = \begin{bmatrix} -2 \\ 4 \\ 0 \\ -6 \end{bmatrix}$

$\because X_1 = -\frac{1}{2} X_2$ i.e $K = -\frac{1}{2}$ (exist)

or $2X_1 + X_2 = 0$ so vectors are LD

eg $X_1 = \begin{bmatrix} 2 \\ 0 \\ 1 \\ -3 \end{bmatrix}$, $X_2 = \begin{bmatrix} -6 \\ -3 \\ -3 \\ 9 \end{bmatrix}$

$\because X_1 \neq KX_2$ for any K

i.e K = DNE

$\Rightarrow$ Vectors are LI

Imp Discussion: (Various Def^n of Rank) →

Def^n ① if $\rho(A_{7\times 6}) = 4 \Rightarrow$
- there exist at least one Non Sing Submatrix of 4×4
- All the submatrices of 5×5 & 6×6 are Singular

Def^n ② if $\rho(A_{7\times 6}) = 4 \Rightarrow$
- A will have at most 4 LI Row vectors
- A will have at most 4 LI Column vectors

Note ① Non Sing $\simeq$ LI & Sing $\simeq$ LD

② If vectors are orthogonal then these are LI also. ($\Rightarrow$ but $\Leftarrow$ ✗)

$1+w^2+w^4$
$1+w^2+w^3+w$
$1+w^2+w$
$= 0$

eg Check the nature of vectors $\begin{bmatrix}2\\-1\\3\\0\end{bmatrix}, \begin{bmatrix}-1\\4\\0\\2\end{bmatrix}, \begin{bmatrix}4\\-9\\3\\-4\end{bmatrix}$

M-I Let the given vectors are

$X_1 = \begin{bmatrix}2\\-1\\3\\0\end{bmatrix}, X_2 = \begin{bmatrix}-1\\4\\0\\2\end{bmatrix}, X_3 = \begin{bmatrix}4\\-9\\3\\-4\end{bmatrix}$

By observation $\begin{bmatrix}2\\-1\\3\\0\end{bmatrix} - 2\begin{bmatrix}-1\\4\\0\\2\end{bmatrix} = \begin{bmatrix}4\\-9\\3\\-4\end{bmatrix}$

i.e. $X_1 - 2X_2 = X_3$

or $\boxed{X_1 - 2X_2 - X_3 = 0}$

M-II Let $A = [X_1\ X_2\ X_3] = \begin{bmatrix}2 & -1 & 4\\-1 & 4 & -9\\3 & 0 & 3\\0 & 2 & -4\end{bmatrix}$ 4×3

$R_2 \leftrightarrow R_1$ $\longrightarrow$ $\begin{bmatrix}\boxed{-1} & 4 & -9\\2 & -1 & 4\\3 & 0 & 3\\0 & 2 & -4\end{bmatrix}$

$R_2 \to R_2 + 2R_1$, $R_3 \to R_3 + 3R_1$ $\longrightarrow$ $\begin{bmatrix}-1 & 4 & -9\\0 & 7 & -14\\0 & 12 & -24\\0 & 2 & -4\end{bmatrix}$

$R_2 \to \frac{1}{7}R_2$, $R_3 \to \frac{1}{12}R_3$, $R_4 \to \frac{1}{2}R_4$ $\longrightarrow$ $\begin{bmatrix}-1 & 4 & -9\\0 & \boxed{1} & -2\\0 & 1 & -2\\0 & 1 & -2\end{bmatrix} \longrightarrow \begin{bmatrix}-1 & 4 & -9\\0 & 1 & -2\\0 & 0 & 0\\0 & 0 & 0\end{bmatrix}$

i.e. $\rho(A) = 2 <$ No of vectors (3)

$\Rightarrow$ Vectors are L.D.

State the vectors $(1\ 2\ -1)^T, (2\ 3\ 0)^T, (4\ -1\ 2)^T, (-2\ 4\ 1)^T$ are ?

Sol: Let the given vectors are $X_1 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, X_2 = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}, X_3 = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}, X_4 = \begin{bmatrix} -2 \\ 4 \\ 1 \end{bmatrix}$

& New Mat is $A = [X_1\ X_2\ X_3\ X_4] = \begin{bmatrix} 1 & 2 & 4 & -2 \\ 2 & 3 & -1 & 4 \\ -1 & 0 & 2 & 1 \end{bmatrix}_{3\times 4}$

$\rho(A) \leq 3$ (By Property of Rank)

ie $\rho(A) < 4$ obviously By Common Reason

or $\rho(A) <$ No. of Vectors $\Rightarrow$ Vectors are L.D

Q: Find λ if there DNE any Relationship b/n the vectors, $\hat{i}+2\hat{j}+3\hat{k}$, $4\hat{i}+\lambda\hat{j}+6\hat{k}$, $3\hat{i}+4\hat{j}+5\hat{k}$ is L.I ?

Sol: given vectors are

$$X_1=\begin{bmatrix}1\\2\\3\end{bmatrix},\ X_2=\begin{bmatrix}4\\\lambda\\6\end{bmatrix},\ X_3=\begin{bmatrix}3\\4\\5\end{bmatrix}$$

Let $A=[X_1X_2X_3]=\begin{bmatrix}1&4&3\\2&\lambda&4\\3&6&5\end{bmatrix}_{3\times3}$

By Tricky Method, For LI vectors

$|A|\neq 0$

$$\begin{vmatrix}1&4&3\\2&\lambda&4\\3&6&5\end{vmatrix}\neq 0$$

$\lambda\neq 5$

it means Vector are L.D. if $\lambda=5$

& Vectors are LI if $\lambda\in R-\{5\}$

GATE 2010 (MS 8) The vectors $(1\ 1\ 1)$ & $(1\ a\ a^2)$ are ? where $a = \frac{-1}{2} + i\frac{\sqrt{3}}{2} = w$

$X_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $X_2 = \begin{bmatrix} 1 \\ a \\ a^2 \end{bmatrix} = \begin{bmatrix} 1 \\ w \\ w^2 \end{bmatrix}$ where $w^3 = 1$ & $1 + w + w^2 = 0$

(a) LD

(b) LI

(c) orthogonal

(d) orthonormal

By observation, $X_1 \neq KX_2$ for any K so X_1 & X_2 are (LI)

Now $X_1 \cdot X_2 = (1)(1) + (1)(w) + (1)(w^2)$

$= 1 + w + w^2 = 0$

$\therefore$ Vectors are orthogonal

Now, $\|X_1\| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$

$\|X_2\| = \sqrt{1^2 + w^2 + (w^3)^2} = 0$

} Not orthonormal

System of Equations

Non Homogeneous System

$AX = B$

$$\left.\begin{aligned} 2x-4y+z&=0\\ x-2+4z&=0\\ 3x+4y-z&=0\\ -7x+2y+3z&=0 \end{aligned}\right\} \quad \left.\begin{aligned} 2x-4y+z&=0\\ x+0y+4z&=2\\ 3x+4y-z&=0\\ -7x+2y+3z&=0 \end{aligned}\right\} \quad \begin{bmatrix} 2 & -4 & 1 \\ 1 & 0 & 4 \\ 3 & 4 & -1 \\ -7 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix}$$

$$A_{4\times 3}\ X_{3\times 1} = B_{4\times 1}$$

where $[A:B] = \left[\begin{array}{ccc:c} 2 & -4 & 1 & 0 \\ 1 & 0 & 4 & 2 \\ 3 & 4 & -1 & 0 \\ -7 & 2 & 3 & 0 \end{array}\right]_{4\times 4}$ = Augmented Mat

Homogeneous System

$AX = O$

① Consider

Coeff. Mat

Constant Mat

$$A_{m\times n}\, X_{n\times 1} = B_{m\times 1}$$

No. of equations

No. of Variables

Solution of System / Variable Mat / unknown vector

② if $m > n$, System is OVER DETERMINED (Tough)

if $m = n$, " " EQUALLY " "

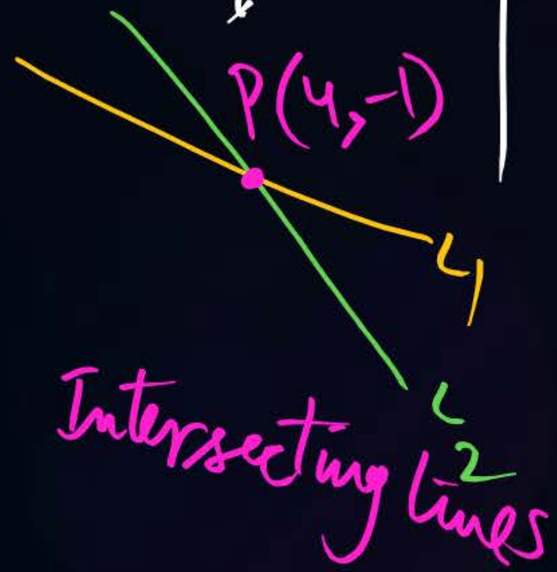
if $m < n$, " " UNDER " " (Easy)

(3) Nature of Solution :-

eg $2x+3y=5$
$x-3y=7$

$X=\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}4\\-1\end{bmatrix}$

= unique sol



Intersecting lines

eg $2x+3y=5$
$4x+6y=10$

$X=\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}1\\1\end{bmatrix},\begin{bmatrix}2\\1/3\end{bmatrix},\begin{bmatrix}1/2\\4/3\end{bmatrix},\begin{bmatrix}-3\\11/3\end{bmatrix}\ldots\ldots$

Infinite sol exist



overlapping lines

eg $\begin{cases}2x+3y=5\\4x+6y=9\end{cases}$

$\begin{cases}2x+3y=5\\2x+3y=4.5\end{cases}$

$X=\begin{bmatrix}x\\y\end{bmatrix}=\begin{bmatrix}?\\?\end{bmatrix}$ = No sol exist



11^{r} lines

METHODS of SOLVING Non Homog. System

$(A_{m\times n}X_{n\times 1} = B_{m\times 1})$

① RANK Method (always applicable)

(i.e. for $m > n$, $m = n$, $m < n$)

(i) if $\boxed{\rho(A) = \rho(A:B)} = $ No. of Variables $\Rightarrow$ unique Sol^n

(ii) if $\boxed{\rho(A) = \rho(A:B)} <$ " " $\Rightarrow$ Infinite Sol^n

(iii) if $\rho(A) \neq \rho(A:B) \Rightarrow$ No Solution.

Matrix Method

(applicable only when $m = n$)

(i) if $|A| \neq 0 \Rightarrow$ unique Sol^n exist

(ii) if $|A| = 0$ & $(\text{adj}\,A)B = 0 \Rightarrow$ (∞ Sol^n.)

(iii) if $|A| = 0$ & $(\text{adj}\,A)B \neq 0 \Rightarrow$ (No Sol^n)

$AX = B \Rightarrow X = A^{-1}B = \dfrac{(\text{adj}\,A)B}{|A|}$

(*) **Consistent System** → if System has Sol (whether unique or Infinite Sol) then System is called Consistent.

Inconsistent System → If System has NO Sol then System is Inconsistent.

(*) Necessary Condition for a System $AX=B$ to be Consistent is $\rho(A)=\rho(A:B)$

OR $\rho(\text{Coeff}^n \text{ Mat}) = \rho(\text{Augmented Mat})$

Qe write the Condition for the existence of at least one Sol of an unknown vector in the System $PY=\phi$

(Consistent System)

$\Downarrow$

$\rho(P)=\rho(P:\phi)$

Sol of System / unknown vector

(*) $\rho(A) \neq \rho(A:B)$ $\Rightarrow$ Rank of Coeff Mat can never exceed Rank of Augmented Mat.

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$\rho(A) = 2 = \rho(A:B)$ (∞ Sol^n)

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & 5 & : & 0 \end{bmatrix}$$

$\rho(A) = 3 = \rho(A:B)$ (Unique Sol^n)

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & 0 & : & 5 \end{bmatrix}$$

$\rho(A) = 2$, $\rho(A:B) = 3$ (No Sol^n)

$$\begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & 5 & : & 14 \end{bmatrix}$$

(Unique Sol^n)

$\rho(A) = 3 = \rho(A:B)$

Ex: The solution of the system

$$x+y+z=6$$
$$x+2y+3z=10$$
$$x+2y+5z=14$$

(a) $\begin{bmatrix}4\\2\\0\end{bmatrix}$ (b) $\begin{bmatrix}2\\4\\0\end{bmatrix}$ (c) ✓ $\begin{bmatrix}4\\0\\2\end{bmatrix}$ (d) $\begin{bmatrix}2\\0\\4\end{bmatrix}$

(M-I) By using options: Ans is (c)

(M-II) w/o using options:

Do yourself as discussed in Class 8th

(M-III) (Using Rank Method) →

$$[A:B]=\left[\begin{array}{ccc:c}1&1&1&6\\1&2&3&10\\1&2&5&14\end{array}\right]\longrightarrow\left[\begin{array}{ccc:c}1&1&1&6\\0&1&2&4\\0&1&4&8\end{array}\right]$$

$$=\left[\begin{array}{ccc:c}1&1&1&6\\0&1&2&4\\0&0&2&4\end{array}\right]_{3\times4}$$

$\rho(A)=3$

$\rho(A:B)=3$

i.e. $\rho(A)=\rho(A:B)=$ No. of Variables (3)

$\Rightarrow$ unique sol will exist.

Process of finding Sol^n

$$AX = B$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 14 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} (x+y+z) \\ (y+2z) \\ 2z \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 4 \end{bmatrix}$$

$$\left.\begin{aligned} x+y+z &= 6 \\ y+2z &= 4 \\ 2z &= 4 \end{aligned}\right\} \Rightarrow z = 2$$

$\& \; y+2(2)=4 \Rightarrow y=0$

also, $x+y+z=6$

$x+0+2=6 \Rightarrow x=4$

i.e. $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$ = unique sol^n.

this method is called RANK Method,

/ Gauss Elimination Method

/ Backward Substitution Method.

Q: Find λ & μ for which following system has ① Infinite sol ② No sol ③ unique sol.

$$x+y+z=6$$
$$x+2y+3z=10$$
$$x+2y+\lambda z=\mu$$

Sol: $$[A:B]=\left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right]$$

$$[A:B]=\left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & (\lambda-3) & (\mu-10) \end{array}\right]_{3\times 4}$$

① for Infinite sol; $\rho(A)=\rho(A:B)<$ No of Variab (3)

Let $\rho(A)=2$ & $\rho(A:B)=2$

$\Downarrow$ $\Downarrow$

$\lambda=3$ $\quad$ $\mu=10$ Ans

② for No sol; $\rho(A)\neq\rho(A:B)\Rightarrow\rho(A)<\rho(A:B)$

Let $\rho(A)=2$ & $\rho(A:B)=3$

$\Downarrow$ $\Downarrow$

$\lambda=3$ & $\mu\neq10$ Ans

③ For unique soln:

$$\rho(A) = \rho(A:B) = \text{No. of Variables } (3)$$

$\rho(A) = 3$, $\rho(A:B) = 3$

$\Downarrow$ $\Downarrow$

$\lambda \neq 3$, μ can take any value.

$\Rightarrow$ $\boxed{\lambda \neq 3 \ \& \ \mu \in R}$ Ans

Q: Which of the following can be taken as condition for Inconsistency

(a) $S(A) = S(A:B)$

(b) $S(A) < S(A:B)$

(c) $S(A) > S(A:B)$

(d) Both (b) & (c)

THANK - YOU